ARJUN RAJESH

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PROFESSIONAL SUMMARY

Third-year Computer Science student with strong foundation in systems programming, software development, and machine learning. Built 15+ production-ready projects including diagnostic tools, custom Linux distributions, ML pipelines, and mobile applications. Proficient in C, C++, Rust, Python with expertise in low-level systems programming and algorithm optimization. Achieved measurable results: 70% efficiency improvements, 3.2x speedup in heterogeneous computing, 98% reduction in diagnostic time. Seeking Software Engineering Internship or Developer role.

EDUCATION

Bachelor of Technology in Computer Science and Engineering — *Aug 2023–May 2027*
Amrita Vishwa Vidyapeetham, Kerala, India — *Currently in 3rd Year*
Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Organization & Architecture, Software Engineering

Higher Secondary Education (Class 12) — *2022*
Kendriya Vidyalaya, Payyanur — CBSE: 76.4% — **JEE Main: 90th percentile**

TECHNICAL SKILLS

Programming: C, C++, Rust, Python, Java, C#, JavaScript, Dart, SQL, Bash, PowerShell
Development: Object-oriented programming, data structures, algorithms, design patterns, debugging, Git/GitHub, agile development, RESTful APIs
Web & Mobile: HTML, CSS, JavaScript, React, Node.js, Flutter, Android SDK, Firebase
Systems: Windows internals, Linux (Arch, Ubuntu, Debian), kernel optimization, system architecture, virtualization (Proxmox), storage systems (TrueNAS, RAID), server administration
Database: SQL, SQLite, Firebase, object-relational mapping (ORM), database design
Machine Learning: Python (NumPy, pandas, scikit-learn), PyTorch, TensorFlow, graph neural networks
Tools: VS Code, Visual Studio, Git, Postman, Jupyter Notebooks, gdb, perf

TECHNICAL PROJECTS

WinKnight — Windows Diagnostics & Self-Healing Toolkit — *Jan 2024–Present*
C#, PowerShell, Windows API — github.com/ARJUN-RAJESH-24/WinKnight

- Architected diagnostic system with 4 modules processing 100+ checks, automatically resolving 85% of Windows issues
- Integrated 5+ Windows APIs (WMI, Event Log, VSS) using OOP, reducing diagnostic time from 2 hours to 2 minutes (98% improvement)
- Built automated restore point management saving 15GB average storage per system

Arcade OS — Custom Arch Linux Gaming Distribution — *Sep 2023–Mar 2024*
Bash, Python, C, Linux — github.com/ARJUN-RAJESH-24/Arcade-OS

- Created 850MB bootable Linux distribution supporting 12+ emulators and 8 gaming platforms with web-based ROM manager
- Optimized kernel achieving 25% reduction in input latency and 40% memory savings through zram compression
- Automated installation scripts reducing setup time from 2 hours to 15 minutes; validated by 50+ users

Material Discovery with AI — Graph Neural Networks — *Mar 2024–Jun 2024*
Python, PyTorch, PyG — github.com/ARJUN-RAJESH-24/Material-Discovery-with-AI

- Built ML pipeline processing 75,000+ crystal structures from Materials Project database for property prediction
- Trained CGCNN achieving 0.92 R^2 score, outperforming baseline models by 18%
- Optimized hyperparameters reducing training time by 35% while improving accuracy by 12%

ML Text Classification System — *Nov 2023–Feb 2024*
Python, scikit-learn, XGBoost, NLP — github.com/ARJUN-RAJESH-24/Machine-Learning-Case-Study

- Designed evaluation framework analyzing 50,000+ text samples across 4 datasets for content moderation
- Implemented TF-IDF vectorization with 100,000+ features; achieved 89% F1-score, outperforming SVM by 12%
- Created automated cross-validation pipeline reducing testing time by 75%

Demon Mode Protocol — Fitness Tracking Mobile App — *Jul 2024–Oct 2024*
Flutter, Dart, Firebase, SQLite — github.com/ARJUN-RAJESH-24/DemonModeProtocol

- Developed Android app with GPS tracking processing 1,000+ location points per session with 95% accuracy
- Engineered offline-first SQLite architecture enabling zero data loss; serving 100+ beta users
- Implemented battery-optimized services consuming 15% less power than alternatives

TECHNICAL EXPERIENCE

Independent Developer & Technical Researcher *Aug 2023–Present*

- Built 15+ system utilities across Windows and Linux, reducing troubleshooting time by 70%
- Deployed home server infrastructure using TrueNAS and Proxmox with RAID configuration from recycled hardware, achieving 99.5% uptime and 5TB storage for media and content
- Benchmarked CPU performance across 6 architectures (AMD Zen, Intel Core) using stress-ng and perf, documenting 25+ optimization strategies
- Prototyped heterogeneous computing workflows (CPU/GPU/NPU) using CUDA, achieving 3.2x speedup
- Tested 50+ Windows applications on Linux using Proton/Wine, achieving 92% compatibility rate

CERTIFICATIONS

Cisco CCST — IT Support & Cybersecurity (2024) — Microsoft/LinkedIn — Software Development Essentials (2024) — Oracle Academy — Java Fundamentals & Advanced Java (2023) — HackerRank: Java & Python (5-star) — ISRO IIRS — AI/ML for Geodata Analysis (2024)

LEADERSHIP & ACTIVITIES

Community Contributions

- Contributed to Indian Sign Language Awareness documentary (5,000+ viewers)
- Volunteer: Amritavarsham 70 Clean-up Drive, Hope Charitable Trust (5+ hours)
- **Rajyapuraskar Award** — Bharat Scouts & Guides (Top 1% nationally)

ADDITIONAL INFORMATION

Languages: English (fluent), Hindi (fluent), Malayalam (native)
Availability: Seeking 2025 internships or full-time opportunities starting May 2027